

Unit 01: Computer Fundamentals

CONTENTS

Objectives

Introduction

- 1.1 Characteristics of Computers
- 1.2 Evolution of Computers
- 1.3 Computer Generations
- 1.4 Five Basic Operations of Computer
- 1.5 Block Diagram of Computer
- 1.6 Applications of Information Technology (IT) in Various Sectors
- 1.7 Input-Output Devices
- 1.8 Latest Input-Output Devices in Market
- 1.9 Memory System in a Computer
- 1.10 Units of Memory
- 1.11 Classification of Primary and Secondary Memory
- 1.12 Memory Instruction Set
- 1.13 Memory Registers

Summary

Keywords

Self Assessment

Answers for Self Assessment

Review Questions

Further Readings

Objectives

After studying this unit, you will be able to:

- understand the basics of computers
- discuss the computer evolution
- explain all generation of computers
- know about the Number system and its various types.

Introduction

The word “computer” comes from the word “compute”, which means, “to calculate”. People usually refer to a computer to be a calculating device that can perform arithmetic operations at high speed. Even though the primary objective behind inventing the computer was focused on developing the fast-calculating machine but currently, more than eighty percent work done by computers is both non-mathematical or non-numerical. Therefore, merely defining a computer as a calculating device is to ignore over eighty percent of its functions.

More accurately, we can define a computer as a device that operates upon data. Data can be anything like biodata, hence computer is used for shortlisting candidates for recruiting; marks

obtained by students in when used for preparing results; details (name, age, sex, etc.) of passengers when used for or railway reservations: or several different parameters when used for solving scientific problems, etc.

Hence, data comes in various shapes and sizes depending upon the type of computer application. A computer can and retrieves data as and when desired. The fact that computers process data is so fundamental that have started calling it a *data processor*.

A data processor gathers data from various incoming sources, merging (the process of mixing or putting together) them all, sorting (the process of arranging in some of the sequences— ascending or descending) them in the desired order, and finally printing them in the desired format. The activity of processing data using a computer is called *data processing*. Data processing consists of three sub-activities: capturing input data, manipulating the data, and managing output results. As used in data processing, information is data arranged in an order and form that is useful to people receiving it. Hence, *data is a raw material used as input to data processing and information is processed data obtained as an output of data processing*.



Did You Know?

People usually consider a computer to be a calculating device that can perform arithmetic operations at high speed. It is also known as a data processor because it not only computes in the usual sense but also performs other functions with the data.

1.1 Characteristics of Computers

The increasing popularity of computers has proved that it is a very powerful and useful tool. The power and use of this popular tool are mainly due to its following characteristics:

1. **Automatic.** Computers are similar to automatic machines that can work by themselves without human intervention. However, computers being machines cannot start themselves and cannot go out and find their own problems and solutions. Computers need to be instructed using coded instructions specifying exactly how they will do a particular job.
2. **Speed.** A computer is a very fast device that can yield output and perform in a few seconds. In other words, a computer can do in a few minutes what would take a man his entire lifetime. The computer speed is measured in terms of seconds or even milliseconds (10⁻³) but in terms of microseconds (10⁻⁶), nanoseconds (10⁻⁹), and even picoseconds (10⁻¹²). A powerful computer is capable of performing several billion (10⁹) simple arithmetic operations per second.
3. **Accuracy.** In addition to being very fast, computers are very accurate. The accuracy of a computer is consistently high and the degree of its accuracy depends upon its design. A computer performs even calculations with the same accuracy. However, errors can occur in a computer. These errors are mainly due to human rather than technological weaknesses.
4. **Diligence.** Unlike human beings, a computer is free from monotony, tiredness, and lack of concentration. It can continuously work for hours without creating any error and without grumbling. Hence, computers score over human beings in doing the routine type of jobs that require great accuracy. If ten million calculations must be performed, a computer will perform the last one with exactly the same accuracy and speed as the first one.
5. **Versatility.** Versatility is one of the most wonderful things about a computer. One moment it is preparing results of an examination, next moment it is busy preparing electricity bills, and in between, it may be helping an office secretary to trace an important letter in seconds. All in all, a computer performs almost any task, provided the task can be reduced to a finite series of logical steps.
6. **Power of Remembering.** As a human being acquires new knowledge, his/her brain subconsciously selects what it feels to be important and worth retaining in memory. The brain relegates unimportant details to the mind or just forgets them. This is not the case with computers. A computer can store and recall any amount of information because of its secondary storage (a type of detachable memory) capability. It can retain a piece of information as long as a user desires and the user can recall the information whenever required. Even after several years, a user can recall the same information that he/she had stored in the computer several years ago.
7. **No IQ.** A computer possesses no intelligence of its own. It needs to be programmed and told about what to do and in what sequence. Hence, only a user determines—at tasks a computer will perform. A computer cannot take its own decision in this regard.
8. **No Feelings.** Computers are devoid of emotions. They have no feelings and no instincts because they are Machines. Although men have succeeded in building a computer memory, no

computer processes the equivalent of a human heart and soul. Based on our feelings, taste, knowledge, and experience we often make certain judgments in our day-to-day life whereas, computers cannot make judgments on their own. They make judgments based on the instructions given to them in the form of programs that are written by us (human beings).

1.2 Evolution of Computers

Necessity is the mother of invention. The saying holds true for computers too. Computers were invented because of man's search for fast and accurate calculating devices. Basic Pascal invented the first mechanical adding machine in 1642. Later, in the year 1671, Baron Gottfried Wilhelm von Leibniz of Germany invented the first calculator for multiplication. Keyboard machines originated States around 1880 and we use them even today. Around the same period, Herman Hollerith came up with the concept of punched cards that were extensively used as an input medium in computers even by the late 1970s. Machines and calculators made their appearance in Europe and America towards the end of the century.

Charles Babbage, a nineteenth-century Professor at Cambridge University, is considered the father of modern digital computers. He had employed a group of clerks for preparing mathematical and statistical tables. Babbage had to spend several hours checking these tables because even utmost care and precautions could not eliminate human errors. Soon he became dissatisfied and exasperated with this type of monotonous job. As a result, he started thinking about building a machine that could compute tables guaranteed to be error-free.

In this process, Babbage designed a "Difference Engine" in the year 1822 that could produce reliable tables. In 1842, Babbage came out with his new idea of a completely automatic Analytical Engine for performing basic arithmetic functions for any mathematical problem at an average speed of 60 additions per minute. Unfortunately, he was unable to produce a working model of this machine because the precision engineering required to manufacture the machine was not available during that period. However, his efforts established several principles that are fundamental to the design of any digital computer. Some of the well-known early computers are as follows:

- **The Mark I Computer (1937-44):** Also known as Automatic Sequence Controlled Calculator, this was the first fully automatic calculating machine designed by Howard A. Aiken of Harvard University in collaboration with IBM (International Business Machines) Corporation. It was an electromechanical device (used both electronic and mechanical components) based on the techniques already developed for punched card machines.

Although this machine proved to be extremely reliable, it was very complex in design and huge in size. It used over 3000 electrically actuated switches to control its operations and was approximately feet long and 8 feet high. It could perform five basic arithmetic operations: addition, subtraction, multiplication, division, and table reference on numbers as big as 23 decimal digits. It took approximately 0.3 seconds to add two numbers and 4.5 seconds for the multiplication of two numbers.

- **The Atanasoff-Berry Computer (1939-42):** Dr. John Atanasoff developed an electronic machine to solve certain mathematical equations. The machine was called the Atanasoff-Berry Computer, or ABC, after its inventor's name and his assistant, Clifford Berry. It used 45 vacuum tubes for internal logic and capacitors for storage.
- **The ENIAC (1943-46).** The Electronic Numerical Integrator And Calculator (ENIAC) was the first all-electronic computer. It was constructed at the Moore School of Engineering of the University of Pennsylvania, U.S.A. by a design team led by Professors J. Presper Eckert and John Mauchly. The team developed ENIAC because of military needs. It was used for many years to solve ballistic-related problems. It took up wall space in a 20 x 40 square feet room and used 18,000 vacuum tubes it could add two numbers in 200 microseconds and multiply them in 2000 microseconds.

- **The EDVAC (1946-52):** A major drawback of ENIAC was that its programs were wired on boards that made it difficult to change the programs. Dr. John Von Neumann later introduced the “stored program” concept that helped in overcoming this problem. The basic idea behind this concept is that a sequence of instructions and data can be stored in the memory of a computer for automatically directing the flow of operations. This feature considerably influenced the development of modern digital computers because of the ease with which different programs can be loaded and executed on the same computer. Due to this feature, we often refer to modern digital computers as stored-program digital computers. The Electronic Discrete Variable Automatic Computer (EDVAC) used the stored’ program concept in its design. Von Neumann also has a share of the credit for introducing the idea of storing both instructions and data in binary form (a system that uses only two digits—0 and 1 to represent all characters), instead of decimal numbers or human-readable words.
- **The EDSAC (1947-49):** Almost simultaneously with EDVAC of the U.S.A., the Britishers developed the Electronic Delay Storage Automatic Calculator (EDSAC). The machine executed its first program in May 1949. In this machine, addition operations took 1500 microseconds, and multiplication operation: took 4000 microseconds. A group of scientists headed by Professor Maurice Wilkes at the Cambridge University Mathematical Laboratory developed this machine.
- **The UNIVAC I (1951):** The Universal Automatic Computer (UNIVAC) was the first digital computer that was not “one of a kind”. Many UNIVAC machines were produced, the first of which was installed in the Census Bureau in 1951 and was used continuously for 10 years. The first business use of a computer, a UNIVAC I, was by General Electric Corporation in 1954. In 1952, the International Business Machines (IBM) Corporation introduced the IBM-701 commercial computer. In rapid succession, improved models of the UNIVAC I and other 700-series machines were introduced. In 1953, IBM produced the IBM-650 and sold over 1000 of these computers. UNIVAC marked the arrival of commercially available digital computers for business and scientific applications.



Did You Know?

Some of the well-known early computers are the MARK 1 (1937-44), the ATANASOFF-BERRY (1939-42), the ENIAC (1943-46), the EDVAC (1946-52), the EDSAC (1947-49), and the UNIVAC I (1951).

1.3 Computer Generations

Generation in computer talk is a step in technology. It provides a framework for the growth of the computer industry. Originally, the term “generation” was used to distinguish between varying hardware technologies but it has now been extended to include both hardware and software that together make up a computer system. The custom of referring to the computer era in terms of generations came into wide use only after 1964. There are 5 computer generations known till today.

First Generation (1942-1955)

We have already discussed some of the early computers—ENIAC, EDVAC, EDSAC, UNIVAC 1, and IBM 701. These machines and others of their time used thousands of vacuum tubes. A vacuum tube [see Figure 1 (a)] was a fragile glass device, which used filaments as a source of electronics and could control and amplify electronic signals. It was the only high-speed electronic switching device

available in those days. These vacuum tube computers could perform computations in milliseconds and were referred to as first-generation computers.

The memory of these computers used electromagnetic relays, and all data and instructions were fed into the system from punched cards. The instructions were written in machine and assembly languages because high-level programming languages were introduced much later. Since machine and assembly languages are very difficult to work with, only a few specialists understood how to program these early computers.

Characteristic features of first-generation computers are as follows:

1. They were the fastest calculating devices of their time.
2. They were too bulky in size, requiring large rooms for installation.
3. They used thousands of vacuum tubes that emitted a large amount of heat and burnt out frequently. Hence, the rooms/areas in which these computers were located had to be properly air-conditioned.
4. Each vacuum tube consumed about half a watt of power. Since a computer typically used more than ten thousand vacuum tubes, the power consumption of these computers was very high.
5. As vacuum tubes used filaments, they had a limited life. Because a computer used thousands of vacuum tubes, these computers were prone to frequent hardware failures.
6. Due to the low mean time between failures, these computers required constant maintenance.
7. In these computers, thousands of individual components were assembled manually into electronic circuits. Hence, the commercial production of these computers was difficult and costly.
8. Since these computers were difficult to program and use, they had limited commercial use.



Task: Give a brief description of first-generation computers.

Second Generation (1955-1964)

John Bardeen, Willian Shockley, and Walter Brattain invented a new electronic switching device called transistor [see Figure 1(b)] at Bell Laboratories in 1947. Transistors soon proved to be a better electronic switching device than vacuum tubes due to their following properties:

1. They were more rugged and easier to handle than tubes since they were made of germanium semiconductor material rather than glass.
2. They were highly reliable as compared to tubes since they had no parts like a filament that could burn out.
3. They could switch much faster (almost ten times faster) than tubes. Hence, switching circuits made of transistors could operate much faster than their counterparts made of tubes.
4. They consumed almost one-tenth of the power consumed by tube and were much smaller than a tube and less expensive to produce as well.
5. They dissipated much less heat as compared to vacuum tubes.

Due to the properties of transistors listed above, these computers were more powerful, more reliable, less expensive, smaller, and cooler to operate than the first-generation computers. They used magnetic cores for main memory and magnetic disk and tape as secondary storage media. Punched cards were still popular and widely used for preparing and feeding programs and data to these computers.

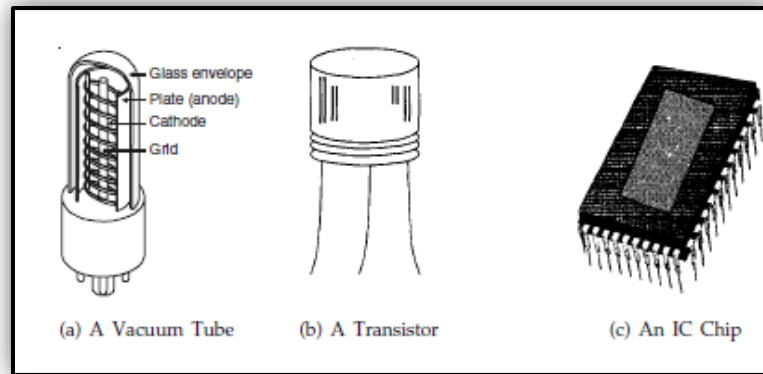


Figure 1: Electronic Devices Used for Manufacturing Computers of Different Generations.

On the software front, high-level programming languages (like FORTRAN, COBOL, ALGOL, and SNOBOL) and batch operating systems emerged during the second-generation. High-level programming languages were easier for people to understand and work with than assembly or machine languages making second-generation computers easier to program and use than first-generation computers. The introduction of batch operating system enabled multiple jobs to be batched together and submitted at a time.

A batch operating system causes an automatic transition from one job to another as soon as the former job completes. This concept helped in reducing human intervention while processing multiple jobs resulting in faster processing, enhanced throughput, and easier operation of second-generation computers.

The first-generation computers were mainly used for scientific computations. However, the second-generation computers were increasingly used in business and industry for commercial data processing applications like payroll, inventory control, marketing, and production planning. The characteristic features of second-generation computers are as follows:

1. They were more than ten times faster requiring smaller space as compared to the first-generation computers.
2. They consumed less power and dissipated less heat than the first-generation computers. The rooms/areas in which the second-generation computers were located still required to be properly air-conditioned.
3. They were more reliable and less prone to hardware failures than the first-generation computers.
4. They had faster and larger primary and secondary storage as compared to first-generation computers.

Such systems were easier to program & were commercially more viable with thousands of individual transistors assembled manually into electronic circuits.



Task

Discuss the second-generation computers.

Third Generation (1964-1975)

In 1958, Jack Clair Kilby and Robert Noyce invented the first integrated circuit. *Integrated circuits (ICs)* are circuits consisting of several electronic components like transistors, resistors, and capacitors grown on a single chip of silicon eliminating wired interconnection between components. The IC technology made it possible to integrate a larger number of circuit components into a very small (less than 5 mm square) surface of silicon, known as “chip” [see Figure 1(c)]. Initially, the integrated circuits contained only about ten to twenty components. This technology was named *small-scale integration (SSI)*. Later with the advancement in technology for

manufacturing ICs, it became possible to integrate up to about components on a single chip. This technology came to be known as *medium-scale integration* (MSI).

Computers built using integrated circuits characterized the third generation. Earlier ones used SSI technology and later used MSI technology. Moreover, the parallel advancements in storage technologies allowed the construction of larger magnetic core-based random access memory as well as larger capacity magnetic disks and tapes. Hence, third-generation computers typically had few megabytes (less than 5 Megabytes) of main memory and magnetic disks capable of storing few tens of megabytes of data per disk drive.

On the software front, standardization of high-level programming languages, timesharing operating systems, unbundling of software from hardware, and creation of an independent software industry happened during the third generation. FORTRAN and COBOL were the most popular high-level programming languages in those days. American National Standards Institute (ANSI) standardized them in 1966 and 1968 respectively, and the standardized versions were called ANSI FORTRAN and ANSI COBOL. Some more high-level programming languages were introduced during the third-generation period. Notable among these were PL/1, PASCAL, and BASIC.

We saw that second-generation computers used batch operating systems. In these systems, users had to prepare their data and programs and then submit them to a computer center for processing. The operator at the computer center collected these user jobs and fed them to a computer in batches at scheduled intervals. The respective users then collected their job output from the computer center. The inevitable delay resulting from this batch processing approach was very frustrating to some users, especially programmers, because often they had to wait for days to locate and correct a few program errors. To rectify this situation, John Kemeny and Thomas Kurtz of Dartmouth College introduced the concept of a timesharing operating system.

Timesharing operating system enables multiple users to directly access and share computing resources simultaneously in a manner that each user feels that no one else is using the computer. This is accomplished by using a large number of independent relatively low-speed, on-line terminals connected to the main computer simultaneously. The introduction of the timesharing concept helped in drastically improving the productivity of programmers and made on-line systems feasible, resulting in new on-line applications like airline reservation systems, interactive query systems, etc.

Until 1965, computer manufacturers sold their hardware along with all associated software without any separate charges for software. From the user's standpoint, the software was free. However, the situation changed in 1969 when IBM and other computer manufacturers began to price their hardware and software products separately. This unbundling of software from hardware allowed users to invest only in the software of their need and value. This led to the creation of many new software houses and the beginning of the independent software industry.

Digital Equipment Corporation (DEC) introduced the first commercially available minicomputer, the PDP-8 (Programmed Data Processor), in 1965. It could easily fit in the corner of a room and did not require the attention of a full-time computer operator. It used a timesharing operating system, and several users could access it simultaneously from different locations in the same building.

The characteristic features of third-generation computers are as follows:

1. They were more powerful than second-generation computers capable of performing about one million instructions per second.
2. They consumed less power and dissipated less heat than second-generation computers. The rooms/areas in which third-generation computers were located still required to be properly air-conditioned.
3. They were smaller, reliable, less prone to hardware failures, & required lower maintenance cost.
4. They had faster and larger primary and secondary storage as compared to second-generation computers.
5. They were general-purpose machines suitable for both scientific and commercial applications.

6. Timesharing operating systems allowed interactive usage and simultaneous use of these systems by multiple users, thereby improving the productivity of programmers down the time and cost of program development by several-fold.



Task: Identify and discuss the various systems belonging to third-generation computer systems.

Fourth Generation (1975-1989)

The average of electronic components packed on a silicon chip doubled each year after 1965. This progress soon led to the era of *large-scale integration* (LSI) when it was possible to integrate over 30,000 electronic components on a single chip, followed by *very large-scale integration* (VLSI) when it was possible to integrate about one million electronic components on a single chip. This progress led to a dramatic development— the creation of a microprocessor.

A *microprocessor* contains all circuits needed to perform arithmetic logic and control: functions, the core activities of all computers, on a single chip. Hence, it became possible to build a complete computer with a Microprocessor, a few additional primary storage chips, and other support circuitry. It started a new social revolution—the *personal computer* (PC) revolution. Overnight computers became incredibly compact. They become inexpensive to make, and suddenly it became possible for anyone to own a computer.

During the fourth generation, semiconductor memories replaced magnetic core memories resulting in large random-access memories with very fast access time. On the other hand, hard disks became cheaper, smaller, and larger in capacity. In addition to magnetic tapes, floppy disks became very popular as a portable medium for porting programs and data from one computer system to another. Another significant development during the fourth-generation period was the spread of high-speed computer networking enabling the interconnection of multiple computers to enable them to communicate and share data. Local area networks (LANs) became popular for connecting computers within an organization or within a campus. Similarly, wide area networks (WANs) became popular for connecting computers located at larger distances. This gave rise to networks computers and distributed systems.

On the software front, several new developments emerged to match the new technologies of the fourth generation. For example, several new operating systems were developed for PCs. Notable among these were MS-DOS, MS-windows, and apple's propriety OS. Since PCs were for individuals who were not computer professionals, companies developed graphical user interfaces for making computers more user-friendly (easier to use).

A *graphical user interface* ((UI) provides icons (pictures) and menus (list of choices) that users can select with a mouse. This enables new computer users to learn to use a PC quickly. Several new PC-based applications were also developed to make PCs a powerful tool. Notable among these were powerful word processing packages that allowed easy development of documents, spreadsheet packages that allowed easy manipulation and analysis of data organized in columns and rows, and graphics packages that allowed the easy drawing of pictures and diagrams.

Another very useful concept that became popular during the fourth-generation period was that of multiple windows on a single terminal screen. This feature allowed users to see the current state of several applications simultaneously in separate windows on the same terminal screen. During the fourth-generation period, UNIX operating system and C programming language also became very popular. The characteristic features of fourth-generation computers are as follows:

1. PCs were smaller and cheaper than mainframes or minicomputers of the third generation.
2. Although the fourth-generation mainframes required proper air-conditioning of the rooms/areas in which they were located, no air-conditioning was required for PCs.
3. They were general-purpose affordable machines that consumed less power, were more reliable, and were less prone to hardware failures than third-generation computers requiring negligible maintenance cost.

4. They had faster and larger primary and secondary storage as compared to third-generation computers.
5. Graphical user interface (GUI) enabled new users to quickly learn how to use computers.
6. PC-based applications made PCs a powerful tool for both office and home usage.
7. The network of computers enabled sharing of resources like disks, printers, etc. among multiple computers and their users. They also enabled several new types of applications involving the interaction among computer users at geographically distant locations.



Task: Briefly describe fourth-generation computers.

Fifth Generation (1989-Present)

The trend of further miniaturization of electronic components, a dramatic increase in power of microprocessor-chips, and an increase in capacity of main memory and hard disk continued during the fifth generation. VLSI technology came *ULSI* (*Ultra Large-Scale Integration*) technology in the fifth generation resulting in the production of Microprocessor chips having ten million electronic components. The speed of microprocessors and the size of the main memory and hard disk doubled almost every eighteen months. As a result, many features found in the CPUs of large mainframe systems of third- and fourth-generation systems became part of microprocessor architecture in the fifth generation. This ultimately resulted in the availability of very powerful and compact computers becoming available at cheaper rates and the death of traditional large mainframe systems.

Due to this fast pace of advancement in computer technology, more compact and more powerful computers are being introduced almost every year at more or less the same price or even cheaper. Notable among these are portable notebook computers that give the power of a PC to their users even while traveling, powerful desktop PCs and workstations, powerful servers, and very powerful supercomputers.

Storage technology has also advanced providing larger and larger main memory and disk storage capabilities available in newly introduced systems. During the fifth generation, optical disks have also emerged as a popular portable mass storage *D-ROM* (*Compact Disk - Read Only Memory*).

There was a tremendous outgrowth of computer networks. Communication technologies become faster day-by-day and more and more computers were networked together. This trend resulted in the emergence and popularity of the Internet and associated technologies and applications. The Internet made it possible for computer users sitting across the globe to communicate with each other within minutes by use of electronic mail (known as *e-mail*) facility. A vast ocean of information became readily available to computer users through the World Wide Web (known as WWW). Moreover, several new types of exciting applications like e-commerce, virtual libraries, virtual classrooms, distance education, etc., emerged during the period.

Tremendous processing power and the massive storage capacity of fifth-generation computers also made them a very popular tool for a wide range of multimedia applications that deal with information containing text, graphics animation, audio, and video data.

The characteristic features of fifth-generation computers are as follows:

1. Portable PCs (called notebook computers) are much smaller, handy, and more powerful than PCs of the fourth generation allowing users to use computing facilities even while traveling.
2. Although fifth-generation mainframes require proper air-conditioning of the rooms/areas in which they are located, no air-conditioning is normally required for notebook computers, desktop PCs, and workstations.
3. They are general-purpose machines that consume less power than their predecessors, are more reliable, and less prone to hardware failures thus requiring negligible maintenance cost.

4. Many of the large-scale fifth-generation systems have a hot-plug feature that enables a failed component to be replaced with a new one without the need to shut down the system. Hence, the uptime of these systems is very high.
5. They have faster and larger primary and secondary storage as compared to their predecessors.
6. The use of standard high-level programming languages allows programs written for one computer to be easily ported to and executed on another computer.
7. More user-friendly interfaces with multimedia features make the systems easier to learn and use by anyone, including children.
8. Newer and more powerful applications, including multimedia applications, make the systems more useful in every occupation.
9. An explosion in the size of the Internet coupled with Internet-based tools and applications have made these systems influence the life of even common people.

Technological progress in this area is continuing at the same rate even now. The fastest-growth period in the history of computing may be still ahead.



Task: Discuss the various features of fifth-generation computers and different systems belonging to this generation.

1.4 Five Basic Operations of Computer

Certain basic operations are performed by a computer system including (Figure 2):

Inputting: It is the process of entering data and instructions into the computer system. It involves basic operations such as the act of feeding data and conversion into the computer-readable format. Various input devices are used to input data into the computer. Input devices also retrieve information off disks. As the computers work with bits, there should be some mechanism to make data understandable by the CPU (the process is called encoding) and also the information produced by the CPU must be converted to the human-readable form (called decoding). The input unit devices are the ones taking care of the encoding. The devices that can send data directly to the CPU or which do not need to encode it before sending it to the CPU are considered *Direct Entry Input Devices* such as scanners. Devices such as a keyboard that require encoding data so that it is in the form a CPU can understand are *Indirect Entry Input Devices*.

Storing: This involves saving data and instructions to make them readily available for initial or additional processing whenever required.

Processing: It consists of performing arithmetic operations or logical operations on data to convert them into useful information. Majorly, extensive calculations & computations are carried out in this step.

Outputting: The process of producing useful information or results for the user such as a printed report or visual display.

Controlling: The task of directing the manner and sequence in which all of the above 4 operations will be performed is called the operation of the control.

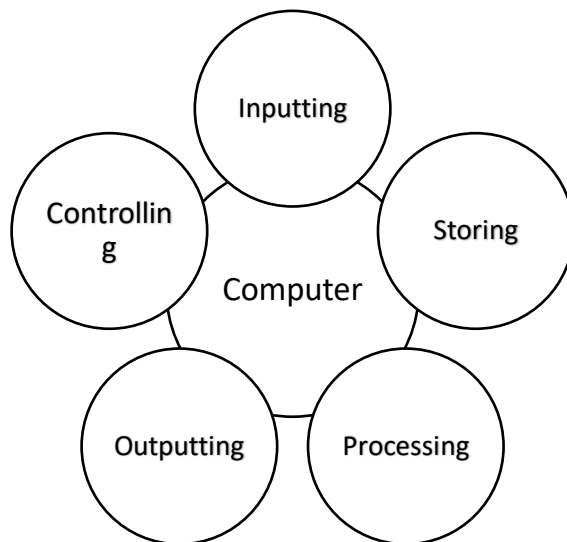


Figure 2: Basic Operations of Computer

1.5 Block Diagram of Computer

The basic components of a computer are:

- Input Unit
- Output Unit
- Memory/Storage Unit
- Arithmetic Logic Unit
- Control Unit
- Central Processing Unit

These are shown in Figure 3.

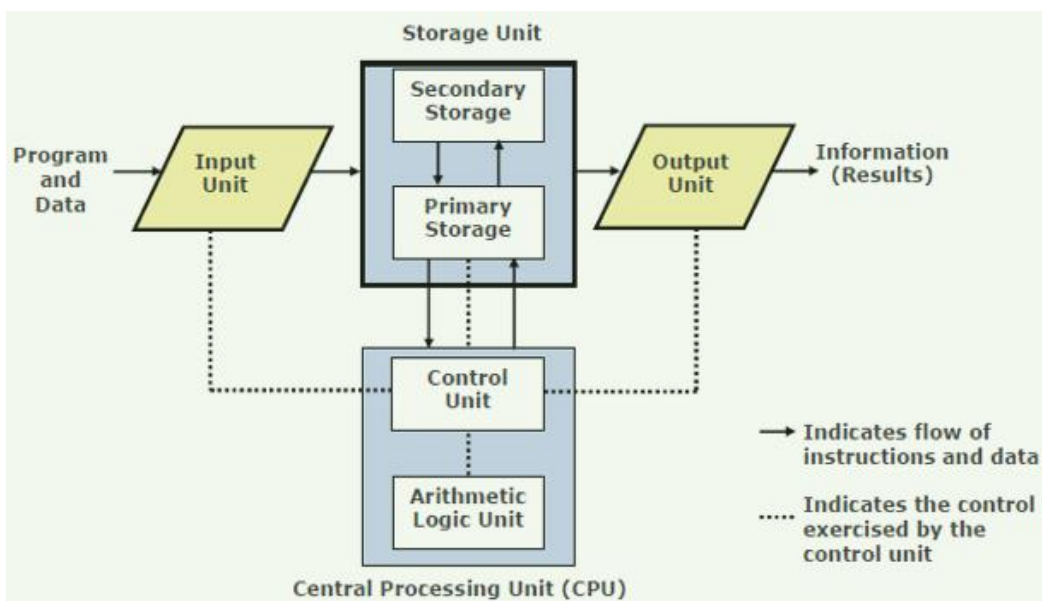


Figure 3: Block Diagram of Computer

A computer can process data, pictures, sound, and graphics. They can solve highly complicated problems quickly and accurately.

Input Unit

Computers need to receive data and instruction to solve any problem. Therefore, we need to input the data and instructions into the computers. The input unit consists of one or more input devices.

The keyboard is one of the most commonly used input devices. Other commonly used input devices are the mouse, floppy disk drive, magnetic tape, etc. All the input devices perform the following functions:

- Accept the data and instructions from the outside world.
- Convert it to a form that the computer can understand.
- Supply the converted data to the computer system for further processing.

Output Unit

The output unit of a computer provides the information and results of a computation to the outside world. Printers, Visual Display Unit (VDU) are the commonly used output devices. Other commonly used output devices are floppy disk drive, hard disk drive, and magnetic tape drive.

Storage Unit

The storage unit of the computer holds data and instructions that are entered through the input unit before they are processed. It preserves the intermediate and final results before these are sent to the output devices. It also saves the data for later use. The various storage devices of a computer system are divided into two categories (Figure 4):

Primary Storage: Stores and provides very fast. This memory is generally used to hold the program being currently executed in the computer, the data being received from the input unit, the intermediate and final results of the program. The primary memory is temporary. The data is lost, when the computer is switched off. To store the data permanently, the data has to be transferred to the secondary memory. The cost of primary storage is more compared to secondary storage. Therefore, most computers have limited primary storage capacity.

Secondary Storage: Secondary storage is used as an archive. It stores several programs, documents, databases, etc. The programs that you run on the computer are first transferred to the primary memory before it is run. Whenever the results are saved, again they get stored in the secondary memory. The secondary memory is slower and cheaper than the primary memory. Some of the commonly used secondary memory devices are Hard disk, CD, etc.

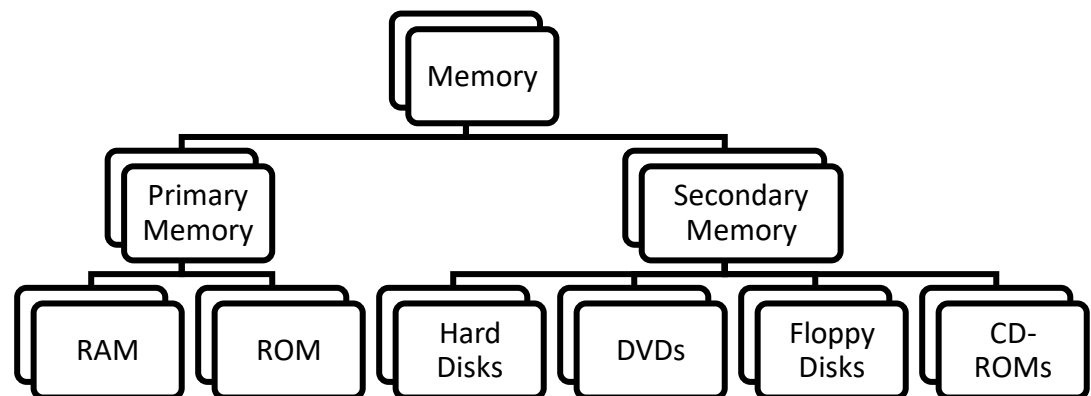


Figure 4: Types of Computer Memory

Arithmetic Logical Unit

All calculations are performed in the Arithmetic Logic Unit (ALU) of the computer. It also does a comparison and takes a decision. The ALU can perform basic operations such as addition, subtraction, multiplication, division, etc, and does logic operations viz, >, <, =, 'etc. Whenever calculations are required, the control unit transfers the data from the storage unit to ALU once the computations are done, the results are transferred to the storage unit by the control unit, and then it is sent to the output unit for displaying results.

Control Unit

It controls all other units in the computer. The control unit instructs the input unit, where to store the data after receiving it from the user. It controls the flow of data and instructions from the storage unit to ALU. It also controls the flow of results from the ALU to the storage unit. The

control unit is generally referred to as the central nervous system of the computer that controls and synchronizes the working.

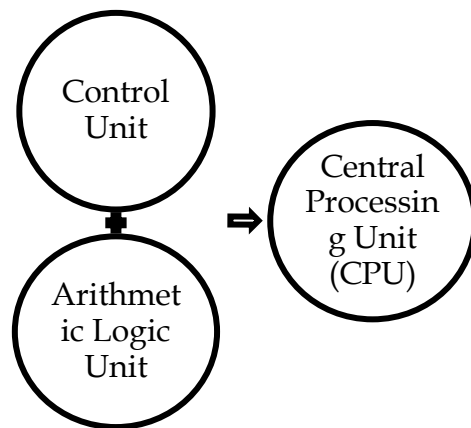


Figure 5: Central Processing Unit (Brain of a Computer System)

Central Processing Unit

The control unit and ALU of the computer are together known as the Central Processing Unit (CPU). The CPU (Figure 5) is like the brain that performs the following functions:

- It performs all calculations.
- It takes all decisions.
- It controls all units of the computer.

1.6 Applications of Information Technology (IT) in Various Sectors

Computers, due to their enormous speed, accuracy, memory, and other invaluable characteristics, are used in all spheres of applications (Figure 6). There is nothing other than computers which have made rapid progress in all spheres.

Computers in Banking: Today, the banking functions in urban and semiurban areas are fully computerized. As a result of the computerization and networking of a large number of banks, several facilities are being offered to the customers. Using online banking, you can check your past transactions from the date of the account opening with a real-time balance.

Money can be easily be transferred to any account across the globe. Internet banking also allows to use of ATM or Debit Card, Credit Card to shop, buy tickets, pay bills for utility services like electricity, telephone, and mobile recharge. You can even subscribe to free monthly bank account statements and apply online to open fixed deposits. the facility of subscribing to mobile banking enables you to virtually carry along the bank with you. Some banking functions carried out by computers are:

- Automated Teller Machines (ATM)
- Online Banking (NEFT, RTGS)
- Core Banking Systems such as Finacle, BaNCs

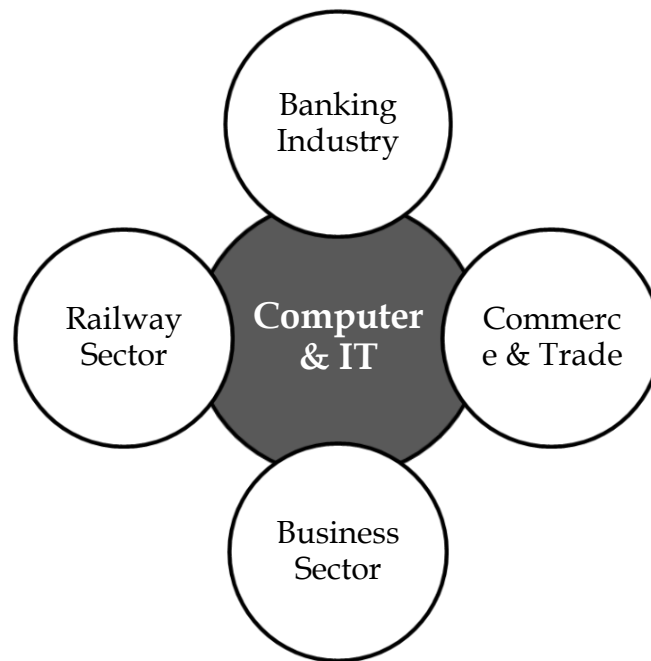


Figure 6: Applications of IT in Various Sectors

Computers in Railway Sector: Railway is the largest organization that is effectively using computers. Computers are being extensively used currently in checking the train records, making online bookings and cancellations, checking train time-tables, live monitoring of the train status, etc.

Computers in Commerce and Trade: The computer is used in all commerce departments like shopping malls and big departmental stores to calculate the sales and purchase of the firm. Computers are very helping in the business to maintain their accounts through the computer. Example: Tally Accounting Software, MARG Software. Nowadays every commercial firm uses computers for every transaction. The ease with which databases, spreadsheets, and data can be compiled on a computer has certainly improved the efficiency and management practices of businesses worldwide. Many offices now use computer programs to handle scheduling, accounting, billing, inventory management, contact management, etc.

Computers in Business: Computer has helped in improving business activities throughout the world. The organization can have access to the latest technology and manpower across the world. Additionally, portable laptop computers, smartphones, wireless internet, air cards, and hub spots are the wave of the future when comes to computer uses in business. Today, the business can be conducted remotely from almost anywhere.

E-commerce has completely changed the scenario of online shopping. Much more online business growth from the application of computer such as Amazon, Flipkart, etc. Global business relations can be now maintained through the Internet. Computers are used in offices for maintaining sales and marketing records, stock control, planning, productions, tax records, and preparing salary sheets, etc.

Computers in Administration: Computers are used for official correspondence, budgeting, accounts, reporting, payroll, attendance, uploading of various schemes, forms, etc. The sector like the census bureau, Income TAX, airline and railway, electricity boards of telephone exchanges are benefiting from computer technology.

Computers in the Healthcare Sector: The application of computers in diagnosing diseases and controlling important medical processes like CTI scan, X-ray, ECG, and Ultrasounds. Computerized equipment helps maintain patients' medical records and perform surgeries. Computers can be used for research and to help doctors in the treatment of diseases. Telehealth, online medicines, telemedicine, telemonitoring are the latest trends in the use of computers in the healthcare sector.

1.7 Input-Output Devices

A computer is a very versatile machine that can easily process different types of data. To work with these data, we require different types of devices. These devices can help us enter data into the computer. These devices are called input and output devices (I/O devices). The devices that are used to display the information of the results are known as output devices. While the devices whose main function is to give instructions and data to the computer are called input devices. The I/O devices are indispensable components that are required for users to communicate with the computer (Figure 7). All in all, the I/O devices act as an intermediary between the user and the machine aiding in the process of communication. These are also known as peripherals since they surround the CPU and memory of a computer system.

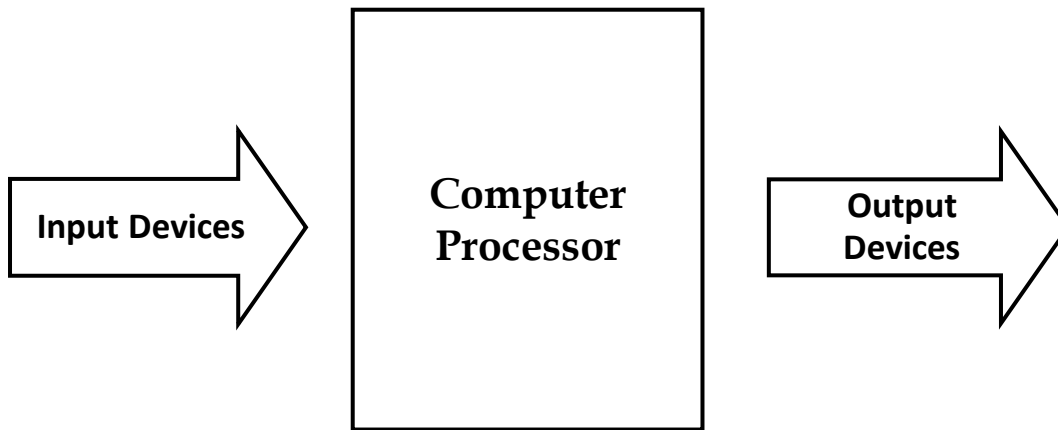


Figure 7: Computers and I/O Devices

Input Devices

Electromechanical devices can be used to enter data and instructions to the computer. Input devices capture and send data and information to a computer. The main function of the input device is to enter the data into the computer. Such devices allow the users to communicate and feed instructions and data to the computers for processing, displaying, storing, and/or transmission. Input devices perform encoding (Converting the human-readable input into the computer-readable format). The various examples of input data are a mouse, keyboard, light pen, etc.

Mouse: The mouse is found on every computer. It is an integral part of it. It is a pointing device (Figure 8). The main principle it works on is the point and click. The mouse uses the trackball to give the motion to the pointer displayed on the screen. Usually, for windows-based programs, a mouse is used. Nowadays, wireless mouse is becoming more popular among people.



Figure 8: Mouse

Keyboard: The keyboard is used to enter all the data into the computer. It directs the computer through instructions. Normally keyboard has 104 buttons called the keys. You can do various tasks through the keyboard. You can program, type, etc. through a keyboard only. The popular keyboard used today is the 101-keys QWERTY keyboard (Figure 9).

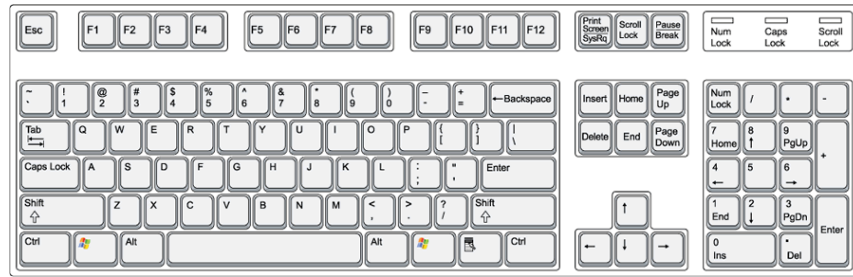


Figure 9: Key Layout on a QWERTY Keyboard

Joystick: Most commonly there are electrical, two-axis joysticks that were invented by C. B. Mirick and name coined by a French pilot Robert Esnault-Pelterie. A joystick is a pointing device that consists of a stick that pivots on a base and reports its angle or direction to the device it is controlling.

Optical Input Devices: Optical input devices use light as a source of input. They eliminate the need for manual entry of data, thus accuracy increases. The most typical types of optical input devices include:

- **Scanners:** A light-sensitive device to copy or capture images, photos, and artwork that exist on paper.
- **Barcode Readers:** An optical scanner that can read bar codes.

Audio Input Devices: The audio-based input devices record the analog sound and convert it into digital form for further processing. There are various types of such devices including voice recognition devices, microphones, musical instruments, etc.

Video-Input Devices: The video input devices are used to enter full-motion video recording into a computer like Cameras.

Output Devices

The computer hardware equipment which gives out the result of the entered input, once it is processed is called an output device. The output devices perform decoding (that is, convert data from machine language to a human-understandable language or convert electronically generated information into human-readable form). These devices are used to communicate the results of data processing carried out by an information processing system. Output devices such as printers, projectors receive the data from a computer usually for display or projection either in softcopy or hard copy format. The different types of output devices include:

Monitor: The monitor is an important component of a computer that is used to display the processed information. It is known as a visual display unit (VDU). Monitors are the computer hardware that displays the video and graphics information generated by the computer as illustrated in Figure 10.



Figure 10: Monitors

The pictures that are displayed on the screen are shown by using a large number of tiny dots known as pixels. These pixels that are shown on the monitor are referred to as the resolution of the screen. There are mostly two types of monitors:

- LCD or liquid crystal display monitor
- CRT or Cathode Ray tube monitor

In modern times, the monitors are typically TFT-LCD or LED display, while older monitors used a CRT display. The LCD monitors are lighter in weight can compare to CRT monitors and they are very popular nowadays. The output that is displayed on the screen is known as the soft copy output as it cannot be retained for a longer time.

Printers: The printer is a very important component of a computer that gives you a printed result of what is displayed on the monitor (Figure 11). The output received from the printer is called hard copy because you can retain it even after you turn off the computer or are no longer working on it. In other words, the printers are the peripherals that are used to print human-readable representations of graphics or text on paper. The world's first computer printer was invented by Charles Babbage. When you look at the printing techniques, there are mainly two types of printers- *Impact printers and Non-impact printers*.



Figure 11: A Printer

The impact printer makes contact with the paper and usually forms the print image by pressing an inked ribbon against the paper using a hammer or pins. The different examples for impact printers are Dot-Matrix, Daisy Wheel. The non-impact printers do not use a striking device to produce characters on the paper; and because these printers do not hammer against the paper that is why they are much quieter. Inkjet and Laser printers are an example of non-impact printers.

Plotters: A plotter is a printer that interprets commands from a computer to make line drawings on paper with one or more automated pens (Figure 12). Unlike a regular printer, the plotter can draw continuous point-to-point lines directly from vector graphics files or commands. The engineering design applications like the architectural plan of a building, design of mechanical components of an aircraft or a car, etc., often require high-quality, perfectly-proportioned graphic output on large sheets. Plotters are an ideal output device for architects, engineers, city planners, and others who need to routinely generate high-precision, hard-copy, the graphic output of widely varying sizes.



Figure 12: A Plotter

The two commonly used types of plotters are drum plotters and flatbed plotters. In a drum plotter, the paper on which the design is to be made is placed over a drum that can rotate in both clockwise and anti-clockwise directions to produce vertical motion. The mechanism also consists of one or more penholders mounted perpendicular to the drum's surface. The pen(s) clamped in the holder(s) can move left to or right to left to produce horizontal motion. A graph-plotting program controls the movements of the drum and pen(s) that is, under computer control, the drum and pen(s) move simultaneously to draw designs and graphs; sheet placed on the drum. The plotter can also annotate the designs and graphs so drawn by using the pen to draw characters of various sizes. Since each pen is program selectable, pens having ink of different colours can be mounted in different holders to produce multi-coloured designs. A flatbed plotter is a computerized plotter that works by using an arm that moves a pen over paper rather than having paper move under the arm as with a drum plotter. The paper is placed on the bed, a flat surface, of the flatbed plotter.

Audio Output Devices: Any devices that attach to a computer to play sound, such as music or speech (speakers, headphones, and sound cards, etc.) are called audio output devices.

- **Speakers:** A device through which we can listen to a sound as an outcome of what we command a computer to do is called a speaker and also are a hardware device which can be attached separately. With the advancement in technology, speakers are now available which are wireless and can be connected using Bluetooth or other applications.
- **Projector:** An optical device that presents an image or moves images onto a projection screen is called a projector. Most commonly these projectors are used in auditoriums and movie theatres for the display of the videos or lighting.
- **Headphones:** A headphone performs the same function as a speaker; the only difference is the frequency of sound. While using the speakers, the sound can be heard over a larger area, and using headphones, the sound is only audible to the person using them. Headphones are also known as earphones or headsets.

1.8 Latest Input-Output Devices in Market

Input Devices:

1. Keyboard
2. Pointing Devices
 - (a) Mouse
 - i. Mechanical mouse
 - ii. Optical mouse
 - (b) Track Ball
 - (c) Joystick
 - (d) Touch Screen
 - (e) Light Pen
 - (f) Touch Pad
3. Digital Camera
4. Scanner
5. Microphone
6. Graphic Digitizer
7. Optical Scanners
8. Optical Character Recognition (OCR)
9. Optical Mark Recognition (OMR)
10. Magnetic-Ink Character Recognition (MICR)



Caution: In the current market there are many local devices are available so never take local devices.



Lab Exercise: Find out the latest trends in the market related to various I/O devices?

1.9 Memory System in a Computer

There are two kinds of computer memory: primary and secondary (Figure 13: Memory Types in a Computer). Primary memory is accessible directly by the processing unit. RAM is an example of primary memory. As soon as the computer is switched off the contents of the primary memory are lost. You can store and retrieve data much faster with primary memory compared to secondary memory. Secondary memory such as floppy disks, magnetic disks, etc., is located outside the computer. Primary memory is more expensive than secondary memory because of this the size of primary memory is less than that of secondary memory.

Computer memory is used to store two things: (i) instructions to execute a program and (ii) data. When the computer is doing any job, the data that have to be processed are stored in the primary memory. This data may come from an input device like a keyboard or from a secondary storage device like a floppy disk.

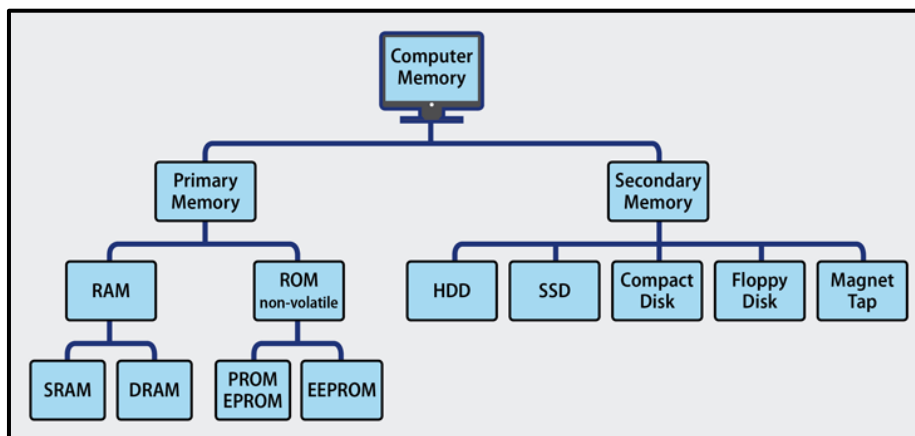


Figure 13: Memory Types in a Computer

Primary Memory

As the program or the set of instructions is kept in primary memory, the computer can follow instantly the set of instructions. In a computer's memory, both programs and data are stored in binary form. The binary system has only two values 0 and 1. These are called bits. As human beings, we all understand the decimal system but the computer can only understand the binary system. It is because a large number of integrated circuits inside the computer can be considered as switches, which can be made ON, or OFF. If a switch is ON it is considered 1 and if it is OFF it is 0. Several switches in different states will give you a message like this: 110101....10.

So, the computer takes input in the form of 0 and 1 and gives output in the form 0 and 1 only. Is it not absurd if the computer gives outputs as 0's & 1's only? But you do not have to worry about it. Every number in a binary system can be converted to a decimal system and vice versa; for example, 1010 meaning decimal 10. Therefore, it is the computer that takes information or data in a decimal form from you, converts it into binary form, processes it producing output in binary form, and again converts the output to decimal form.

The internal storage section is made, up of several small storage locations called cells. Each of these cells can store a fixed number of bits called word length. Each cell has a unique number assigned to it called the address of the cell and it is used to identify the cells. The address starts at 0 and goes up to (n-1). You should know that the memory is like a large cabinet containing as many drawers as there are addresses on memory. Each drawer contains a word and the address is written on the outside of the drawer.

Capacity of Primary Memory

You know that each cell of memory contains one character or 1 byte of data. So, the capacity is defined in terms of byte or words. Thus 64 kilobytes (KB) memory is capable of storing $64 \times 1024 = 32,768$ bytes. (1 kilobyte is 1024 bytes). A memory size ranges from few kilobytes in small systems to several thousand kilobytes in a large mainframe and supercomputer. In your personal computer, you will find memory capacity in the range of 64 MB, 128 MB, 216 MB, and even 1 GB.

The following terms related to the memory of a computer are discussed below:

- **Random Access Memory (RAM):** The primary storage is referred to as random access memory (RAM) because it is possible to randomly select and use any location of the memory to directly store and retrieve data. It takes the same time to any address of the memory as the first address. It is also called read/write memory. The storage of data and instructions inside the primary storage is temporary. It disappears from RAM as soon as the power to the computer is switched off. The memories, which lose their content on the failure of power supply are known as volatile memories. Thus, RAM is called volatile memory.
- **Read-Only Memory (ROM):** Read-only memory (ROM) is a type of non-volatile memory used in computers and other electronic devices (Figure 14). The memory from which we can only read but cannot write on it. In other words, the data held in ROM cannot be electronically modified after the manufacture of the memory device. The information is stored permanently in such memories during manufacture. A ROM stores such instructions that are required to start a computer. This operation is referred to as *bootstrap*. ROM chips are not only used in the computer but also in other electronic items like the washing machine and microwave oven.

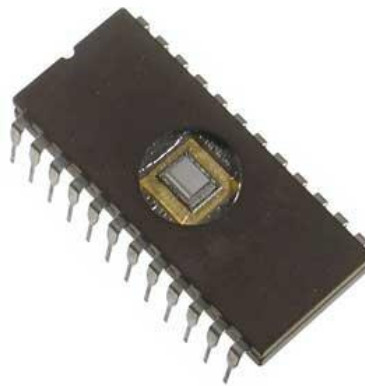


Figure 14: Read-Only Memory

Read-only memory is useful for storing software that is rarely changed during the life of the system, also known as firmware. Software applications (like video games) for programmable devices can be distributed as plug-in cartridges containing ROM.

There are two types of read-only memory (ROM) - manufacturer-programmed and user-programmed. A *manufacturer-programmed ROM* is one in which data is burnt in by the manufacturer of the electronic equipment in which it is used. For example, a personal computer manufacturer may store the system boot program permanently in a ROM chip located on the motherboard of each PC manufactured by it. Similarly, a printer manufacturer may store the printer controller software in a ROM chip located on the circuit board of each manufactured by it. Manufacturer-programmed ROMs are used mainly in those cases where the demand for such programmed ROMs is large. Note that manufacturer-programmed ROM chips are supplied by the manufacturers of electronic equipment and a user can't modify the programs or data stored in a ROM chip.

On the other hand, a *user-programmed ROM* is one in which a user can load and store “read-only” programs and data. That is, it is possible for a user to “customize” a system by converting his/her programs to micro-programs and storing them in a user-programmed ROM chip. Such a ROM is commonly known as *Programmable Read-Only Memory (EPROM)* because a user can program it. Once the user programs are stored in a PROM chip, they can be executed usually in a fraction of the time previously required.

- **Basic Input-Output System (BIOS):** It is a type of ROM that aids in establishing basic communication initially when a computer is turned on.
- **MROM (Masked ROM):** The very first ROMs were hard-wired devices consisting of a pre-programmed set of data or instructions and known as MROMs, which were inexpensive.
- **Programmable ROM:** PROM is read-only memory that can be modified only once by a user. The user buys a blank PROM and enters the desired contents using a PROM program. Inside the PROM chip, there are small fuses that are burnt open during programming. It can be programmed only once and is not erasable.

PROMs are programmed to record information using a special device known as PROM-programmer. However, once the chip has been programmed, the PROM becomes a ROM. That is, the information recorded in it can only be read (it cannot be changed). PROM is also non-volatile storage, i.e., the stored information remains intact even if power is switched off or interrupted.

- **Erasable PROM (EPROM):** EPROM can be erased by exposing it to ultra-violet light for a duration of up to 40 minutes. Usually, an EPROM eraser achieves this function. During programming, an electrical charge is trapped in an insulated gate region. The charge is retained for more than 10 years because the charge has no leakage path. For erasing this charge, ultra-violet light is passed through a quartz crystal window (lid). This exposure to ultra-violet light dissipates the charge.

EPROM chips are of two types – one in which the stored information is erased by exposing the chip for some time of ultraviolet light and the other one in which the stored information is erased by using high voltage electric pulses is known as Ultra-violet EPROM (UVEPROM) and the latter is known as Electrically EPROM (EEPROM). It is easier to alter information stored in an EEPROM chip as compared to a UVEPROM chip. EPROM is also known as *flash memory* because of the ease with which programs stored in it can be altered. Flash memory is used in many new I/O and storage devices; Universal Serial Bus (USB) pen drives and MP3 music players.

- **Electrically Erasable and Programmable Read-Only Memory (EEPROM):** EEPROM is programmed and erased electrically. It can be erased and reprogrammed about ten thousand times. Both erasing and programming take about 4 to 10 millisecond/ms. In EEPROM, any location can be selectively erased and programmed one byte at a time, rather than erasing the entire chip. Hence, the process of reprogramming is flexible but slow.
- **Cache Memory:** Cache memory is commonly used for minimizing the memory-processor speed mismatch because the rate of data fetching by a computer’s CPU from its main memory is about 100 times faster than that from high-speed secondary storage like a disk. However, even with the use of main memory, memory-processor speed mismatch becomes a bottleneck in the speed with which the CPU can process instructions because there is a 1 to 10-speed mismatch between the processor and memory. That is, the rate at which data can be fetched from memory is about 10 times slower than the rate at which the CPU can process

data. The overall performance of a processor can be improved greatly by minimizing the memory processor speed mismatch.

Cache memory acts as a high-speed buffer between CPU and main memory and is used to temporarily store very active data and instructions during processing. Caches are extremely fast, with small memory between CPU and main memory having the access time closer to the processing speed of the CPU. Caches are used to store very active data and instructions during processing temporarily. Since, the cache memory is faster than main memory, processing speed is improved by making the data and instructions needed for current processing available in the cache. Cache memory acts as a high-speed buffer between CPU and main memory and is used to temporarily store very active data and instructions during processing.

- **Virtual Memory:** Virtual memory appears to exist as main storage although most of it is supported by data held in secondary storage, transfer between the two being made automatically as required.
- **Registers:** Registers are a type of computer memory used to quickly accept, store, and transfer data and instructions that are being used immediately by the CPU. The registers used by the CPU are often termed processor registers.



Task: What is the difference between cache memory and RAM?

Secondary Memory: Secondary memory is physically placed or located inside a separate storage device which is then connected to the computer for access. HDD and SSD are the most commonly used secondary memory devices in a computer normally. They are not as expensive as the RAM and ROM and are reasonably priced. Their read and write speed are also comparatively lower. Since it is a separate device altogether, it can preserve the data that is saved inside it without the need for power. The data exchanged and stored is first sent to the RAM and then to the CPU for further processing. They have the possibility of permanently storing the data and at the same time, you can easily use them anywhere and carry them along with you.

1.10 Units of Memory

CPU contains necessary circuitry for data processing controlling other components of the computer. However, one thing it does not have built into it is the place to store programs and data needed during data processing. CPU contains several registers for storing data and instructions but they can store only a few bytes at a time. They are just sufficient to hold only one or two instructions with corresponding data. If the instructions and data of a program being executed by a CPU were to reside in secondary storage like a disk, and fetched and loaded one by one into CPU registers as the program execution proceeded, this would lead to the CPU being idle most of the time. This is because there is a large speed mismatch between the rate at which the CPU can process data and the rate at which data can be transferred from disk to CPU registers. For example, a CPU can process data at a rate of about 5nanosecond/byte and a disk reader can read data at a speed of about 5microseconds/byte. Hence, within the time in which a disk can supply one byte of data, a CPU can process 1000 bytes. This would lead to a very slow overall performance even if a computer used a very fast CPU.

To overcome this problem, there is a need to have a reasonably large storage space that can hold the instructions and data of the program(s) on which CPU is currently working time to fetch and load data from this storage space into CPU registers must also be very small as compared to that space from disk storage to reduce the speed mismatch problem with CPU speed. Every computer has such a storage space known as primary storage, main memory, or simply memory. It is a temporary storage area built into the computer hardware. Instructions and data of a program reside mainly in this area when the CPU is executing the program. Physically, this memory consists of some integrated circuit (IC) chips either on the motherboard or on a small circuit board attached to the motherboard of a computer system. This built-in memory allows the CPU to store and retrieve data

very quickly. The rate of fetching data from this memory is of the order of 50nanoseconds/byte. Hence, the rate of data fetching from main memory is about 100 times faster than that from high-speed secondary storage like a disk.

The memory unit is the amount of data that can be stored in the storage unit. This storage capacity is expressed in terms of Bytes. There are different units of memory as listed in Table 1.

Table 1: Memory Units

Name	Symbol	Binary Measurement	Decimal Measurement	Number of Bytes	Equal to
kilobyte	KB	2^{10}	10^3	1,024	1,024 bytes
megabyte	MB	2^{20}	10^6	1,048,576	1,024 KB
gigabyte	GB	2^{30}	10^9	1,073,741,824	1,024 MB
terabyte	TB	2^{40}	10^{12}	1,099,511,627,776	1,024 GB
petabyte	PB	2^{50}	10^{15}	1,125,899,906,842,624	1,024 TB
exabyte	EB	2^{60}	10^{18}	1,152,921,504,606,846,976	1,024 PB
zettabyte	ZB	2^{70}	10^{21}	1,180,591,620,717,411,303,424	1,024 EB
yottabyte	YB	2^{80}	10^{24}	1,208,925,819,614,629,174,706,176	1,024 ZB



Task: How many bytes in 2 GB describe with example?

1.11 Classification of Primary and Secondary Memory

Computers work with the help of computer memory as they safely keep the data while they have not been processed yet. Many different types of computer memory can be used in different technologies. Computer memory is a generic term for all of the different types of data storage technology that a computer may use, including RAM, ROM, and flash memory. Some types of computer memory are designed to be very fast, meaning that the CPU can access data stored there very quickly. Other types are designed to be very low cost so that large amounts of data can be stored there economically. A computer system is built using a combination of these types of computer memory, and the exact configuration can be optimized to produce the maximum data processing speed or the minimum cost, or some compromise between the two.

Memory is required in computers to store data and instructions. Computer memory is of two basic types – Primary memory and Secondary memory as depicted in Figure 13. We have already discussed the primary and secondary memory above. Let us explore the further subsequent types of primary and secondary memory each.

Primary Memory Types

Prominently, there are two key types of primary memory:

- RAM, or Random-Access Memory

- ROM, or Read-Only Memory

Random Access Memory

Random-access memory (**RAM / r æ m /**) is a form of computer memory that can be read and changed in any order, typically used to store working data and machine code. The data that is stored in the RAM can be accessed randomly. RAM device allows data items to be read or written in almost the same amount of time irrespective of the physical location of data inside the memory. There is no particular order needed. It is the most expensive type of memory. It is very fast and quick but requires power to run and to exchange data and share data that is stored in its memory. Whichever data is needed to be processed imminently is moved to the RAM where the modification and accessing of the data is done quickly so that there is no waiting period for the CPU to start functioning. It improves the speed of the processing of the CPU and the different multiple commands given to it by the users. Whichever data becomes out of use is then move to the secondary memory and the space in the RAM is freed for further processes.

RAM capacity can be enhanced by adding more memory chips on the mother board. The additional RAM chips, which plug into special sockets on the motherboard, are known as *single-in-line memory modules (SIMMs)*. There are two major types of RAM:

a. Dynamic RAM

Dynamic RAM is the commonly used RAM in computers. Unlike earlier times when the computers used to use single data rate (SDR) RAM, now they use dual data rate (DDR) RAM which has faster processing ability. DDR2, DDR3, DDR4 are the available versions of the DRAM each efficient according to their number. The compatibility of the DRAM with the operating system needs to be checked before any installation. The other characteristic features of DRAM are:

- DRAM has memory cells with a paired transistor and capacitor.
- Data is stored in capacitors.
- Capacitors that store data in DRAM gradually discharge energy, no energy means the data has been lost. So, a periodic refresh of power is required to function.
- DRAM is called dynamic as constant change or action i.e. refreshing is needed to keep the data intact.
- It is used to implement main memory.
- Low costs of manufacturing and greater memory capacities.
- Slow access speed and high-power consumption.

b. Static RAM

Static RAM is faster than Dynamic RAM and therefore it is more expensive. It has a total of six transistors in each of its cells. Therefore, it is commonly used as a data cache within the CPU and in high-end servers. It helps in the speed improvements of the computer and possesses the following additional characteristics:

- Uses multiple transistors, typically four to six, for each memory cell but doesn't have a capacitor in each cell. It is used primarily for the cache.
- Data is stored in transistors and requires a constant power flow.
- Because of the continuous power, SRAM doesn't need to be refreshed to remember the data being stored.
- Static as no change or action i.e. refreshing is not needed to keep the data intact. It is used in cache memories.
- Low power consumption and faster access speeds.

SRAM has certain disadvantages associated with it including:

- Fewer memory capacities
- High costs of manufacturing

Table 2 lists the comparison between SRAM and DRAM.

Table 2: Comparison SRAM vs DRAM

S.No.	SRAM	DRAM
1.	Transistors are used to store information in SRAM.	Capacitors are used to store data in DRAM.
2.	Capacitors are not used hence no refreshing is required.	To store information for a longer time, the contents of the capacitor need to be refreshed periodically.
3.	SRAM is faster and expensive	DRAM provides slow access speeds.
4.	SRAMs are low-density devices and used in cache memories	DRAMs are high-density devices and used in main memories.

Read-Only Memory

Data can be only read and not written to it. It is a fast type of memory. It is a non-volatile type of memory. In case of no power, the data is stored in it and then accessed by the computer when power is turned on. It usually contains the Bootstrap code which is required for the computer to interact with the hardware systems and understand its operations and functions according to the commands given to it. In simpler devices used commonly, there is ROM in which the firmware data is stored to get them to function according to their need. ROM is of different types as listed in Table 3.

Table 3: Different Types of ROMs

Type of ROM	Usage
Manufacturer-programmed ROM	Data is burnt by the manufacturer of the electronic equipment in which it is used.
User-programmed ROM / Programmable ROM(PROM)	The user can load and store "read-only" programs and data in It.
Erasable PROM (EPROM)	The user can erase information and the chip can be reprogrammed to store new information.
Ultra Violet EPROM (UVEPROM)	Type of EPROM in which stored information is erased by exposing the chip for some time to the ultra-violet light.
Electrically EPROM or Flash Memory	Type of EPROM in which stored information is erased by using high voltage electric pulses.

Secondary Memory Types

Secondary memory comprises the storage devices that are built into the computer or connected to the computer. They are also known as External Memory or Auxiliary Storage. The computer usually uses its input/output channels to access secondary storage and transfers the desired data using an intermediate area in primary storage. Secondary memory is non-volatile, that is, it retains data when power is switched off and possesses large capacities to the tune of terabytes.

Depending on whether the second memory device is part of the CPU or not, there are two types of secondary memory (Figure 15):

- **Fixed Secondary Memory:** It is an internal media device that is used by a computer system to store data and is usually referred to as Fixed Disks drives or Hard Drives. Literally, the fixed secondary memory is not fixed and can be removed from the system for repairing work, maintenance purposes, and upgrade, etc.
- **Removable Secondary Memory:** The storage device that can be removed/ejected from a computer system while the system is running is called removable secondary memory. It makes it easier for a user to transfer data from one computer system to another and initiates faster data transfer rates. The major examples include HDDs, CD/DVD, Pendrives, SSDs, Memory card, Blue-ray Disk.

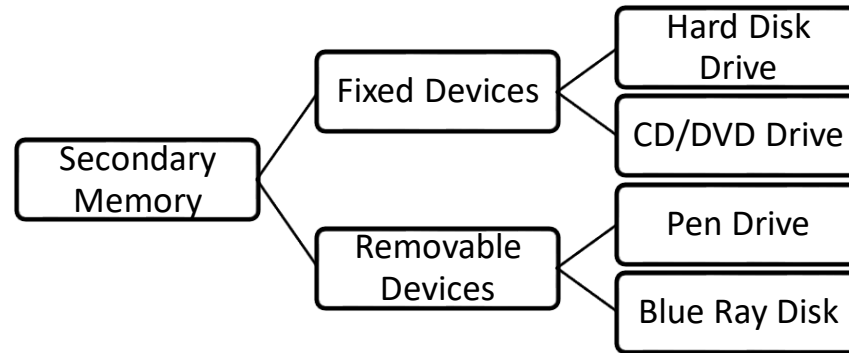


Figure 15: Types of Secondary Memory

Secondary memory comprises many different storage media which can be directly attached to a computer system. These include:

- **Hard Disk Drives:** Hard disk drives are among the most common types of computer memory that can store data and information for a long-term basis. Hard drives work much like records. They are spinning platters that have arms with heads to touch the different parts of it. When the head is at a certain location, data and information is either being written or deleted from that area of the hard drive.

The speed of HDD refers to the speed at which content can be read and written on a hard disk with the set rotation speed varying from 4500 to 7200 rpm (revolutions per minute). The disk access time in HDD is measured in milliseconds. HDDs are of 2 types: Internal and external HDD. Table 4 lists the comparison between both HDD types.

Table 4: Comparison of Internal and External HDD

Features	Internal HDD	External HDD
Portability	No	Yes
Price	Less expensive	More expensive
Speed	Fast	Slow
Size	Big	Small

- **Solid State Drives (SSDs):** SSDs are non-volatile storage devices capable of holding large amounts of data and use NAND flash memories (millions of transistors wired in a series on a circuit board). They provide immediate access to the data as no moving parts are used thus making them much faster and significantly more expensive than the traditional HDDs. The capacity of SSDs is usually measured in Gigabytes. These can be installed inside a computer or purchased in a portable (external) format and are small in physical size, very light, ideal for portable devices.

- **Optical (CD or DVD) Drives:** Optical drives hold content in the digital format and are read using a laser assembly is considered optical media. They use a laser to scan the surface of a spinning disc made from metal and plastic. The disc's surface is divided into tracks, with each track containing many flat areas and hollows. The flat areas are known as lands and the hollows as pits. When the laser shines on the disc's surface, lands reflect the light, whereas pits scatter the laser beam. A sensor looks for the reflected light. The reflected light (lands) represents a binary '1', and no reflection (pits) represents a binary '0'. The most common types of optical media are:

- **Compact Disc (CD):** Compact discs are used to store photos, audio, and computer software and are available in three types which are read-only, recordable, and rewriteable. The different types of CDs are as follows:

Compact Disc Read-Only Memory (CD-ROM):

- The data stored on CD-ROM can only be read. It cannot be deleted or changed.
- Store up to 700MB of data.
- Mostly used to store photos and audios. It is often used to distribute new application software and games.

CD-Recordable (CD-R)

- The user can write data on CD-R only once but can read it many times.
- The data written on CD-R cannot be erased.
- CD-R drives are known as CD burners.
- The process of recording data on CD-R is called burning. CD-R is known as WORM (Write Once Read Many).

Compact Disc Rewriteable (CD-RW)

- Also known as an erasable optical disc.
- The most common type of erasable and rewritable optical disc is the magneto-optical disc.

- **Digital Versatile Disc (DVD):** These are also called Digital Video Discs having storage capability of up to 17 GB of data, much greater than the CD. DVDs are available in three types which are read-only, recordable, and rewritable. Digital Video Disc Read-Only Memory (DVD-ROM) is a high-capacity optical disc that the users can only read but not write or erase. Digital Video Disc Recordable (DVD-R) is similar to a CD-R disc. The written data cannot be erased. Digital Video Disc Rewritable (DVD-RW) enables the users to write data on CD-RW many times by erasing the existing contents.

- **Blu-ray (BD):** Blu-Ray disc is a new and more expensive DVD format that provides higher capacity and better quality than standards DVDs especially for high-definition video. It can store up to 100 GB of data.

- **Tape drives:** Tape drives are types of computer memory that work like tapes or audio cassettes. They have ribbons and spools, where the data and information you would like to store are saved with the ribbon's polarity. These tape drives are mostly used by medium to large organizations as they offer long-term and stable storage.
- **USB Flash Drive:** USB flash drives are small, portable flash memory cards that plug into a computer's USB port and function as a portable hard drive. They are available in sizes such as 256MB, 512MB, 1GB, 5GB, 16GB, and 32 GB are an easy way to transfer and store information.
- **Memory Card:** An electronic flash memory Storage Disk (SD) commonly used in consumer electronic devices such as digital cameras, MP3 players, mobile phones, and other small portable devices. These can be read by connecting the device containing the card to your computer, or by using a USB card reader.

1.12 Memory Instruction Set

Every central processing unit is having a unique instruction set that is distinct to it and identifies a particular processor family. Every CPU has a built-in ability to execute a particular set of machine instructions. So, a computer operates on this different set of instructions over there. Basically, every ALU (the main part that is executing all the computations and calculations) also operates on a certain set of instructions and every CPU has some built-in instruction set that is a particular set of machine instructions that is called as is instruction set.

Most of the CPUs have around 200 or more instructions, such as addition subtraction comparison, division, multiplication, in their instruction set. So, when we want to divide a particular CPU, basically different manufacturers, what they do they have different CPUs that are designed by them. Suppose, a particular CPU belongs to one family and another CPU belongs to another family, a third CPU belongs to another family, and so on. If we want to identify each family which is in turn identified based on the instruction set of the CPU. Thereby, the manufacturers group up to their CPUs into different families that are having similar instruction sets.

New CPU whose instruction set includes instruction set of its predecessor CPU is said to be backward compatible with its predecessor. When any new CPU is generated that is having some instruction set, if its instruction set is quite similar or includes the instruction set of its predecessor, then, in that case, we say that the new CPU is backward compatible with its predecessor. So, it is the backward compatibility of a particular CPU.

1.13 Memory Registers

Registers are a type of computer memory used to quickly accept, store, and transfer data and instructions that are being used immediately by the CPU. Registers are the special memory units that hold information temporarily as the instructions are interpreted and executed by the CPU. Actually, the registers are part of the CPU (not the main memory). The registers have variable word size which signifies the length of a register equaling the number of bits it can store. Different types of registers are used in computer systems. Table 5 lists the various registers.

Table 5: Most Common Registers Used in a Basic Computer

S. No.	Register	Function
1.	Memory Address (MAR)	Holds the address of the active memory location.
2.	Memory Buffer (MBR)	Holds contents of the accessed (read/written) memory word.
3.	Program Control (PC)	Holds the address of the next instruction to be executed.
4.	Accumulator (A)	Holds data to be operated upon, intermediate results, and final results.
5.	Instruction (I)	Holds an instruction while it is being executed.
6.	Input/ Output (I/O)	Used to communicate with the I/O devices.

Output Devices:

1. Monitor
2. Compact Disk
3. Printers
4. Speaker
5. Disk Drives
6. Floppy Disk

7. Headphone

8. Projector

Summary

Computer possesses various characteristics including Automatic Machine, Speed, Accuracy, Diligence, Versatility, Power of Remembering.

Computer generation like First Generation (1942-1955), Second Generation (1955-1964), Third Generation (1964-1975), Fourth Generation (1975-1989), and Fifth Generation (1989-Present).

Block Diagram of Computer means subpart of computers like input, output, and memory devices.

CPU is a combination of ALU and AU.

Secondary storage is a hard disk that saves more data on the computer.

Input devices provide the input from the user side in the computer while the output devices display the results of the computer.

Non-impact printers are larger but are very quiet in operation. Being of non-impact type, they cannot be used to produce multiple copies of a document in a single printing.

Keywords

Data processing: The activity of processing data using a computer is called data processing.

Generation: Originally, the term “generation” was used to distinguish between varying hardware technologies but it has now been extended to include both hardware and software that together make up a computer system.

Integrated circuits: They are usually called ICs or chips. They are complex circuits that have been etched onto tiny chips of semiconductor (silicon). The chip is packaged in a plastic holder with pins spaced on a 0.1" (2.54 mm) grid which will fit the holes on stripboard and breadboards. Very fine wires inside the package link the chip to the pins.

Storage unit: The storage unit of the computer holds data and instructions that are entered through the input unit before they are processed. It preserves the intermediate and final results before these are sent to the output devices.

Flash memory: Storage technology Recall that flash is a non-volatile, Electrically Erasable Programmable Read-Only Memory (EEPROM) chip.

Plotter: Plotters are an ideal output device for architects, engineers, city planners, and others who need to routinely generate high-precision, hard-copy, the graphic output of widely varying sizes.

LCD: LCD stands for Liquid Crystal Display, referring to the technology behind these popular flat-panel monitors.



Lab Exercise: Explain the various computer generations while listing the main features of each.

Self Assessment

1. Which of the following is not a Primary Storage device?
 - A. RAM
 - B. Graphics Card
 - C. ROM
 - D. USB

2. Which of the following is NOT an input device?
 - A. Barcode Reader
 - B. Scanner
 - C. Touchpad
 - D. Speaker

3. A computer is accurate, but if the result of a computation is false, what is the main reason for it?
 - A. Power failure
 - B. The computer circuits
 - C. Incorrect data entry
 - D. Distraction

4. The additional RAM chips which plug into special sockets on the motherboard are known as
 - A. SIMMs
 - B. SIMs
 - C. PROM
 - D. All of these

5. Which of the following belongs to Fifth generation computers?
 - A. Laptop
 - B. NoteBook
 - C. UltraBook
 - D. All of the above

6. Memory is
 - A. A device that performs a sequence of operations specified by the instructions in memory
 - B. The device where information is stored
 - C. a sequence of instructions
 - D. typically characterised by interactive processing and time-sharing of the CPU's time to allow quick response to each user.

7. The QWERTY keyboard
 - A. is the most popular keyboard
 - B. is the fastest keyboard
 - C. is a keyboard rarely used
 - D. uses Dvorak layout

Answers for Self Assessment

- | | | | | |
|------|------|------|------|------|
| 1. D | 2. C | 3. C | 4. A | 5. D |
| 6. B | 7. A | | | |

Review Questions

1. Define Primary memory? Explain the difference between RAM and ROM?
2. Discuss the block structure of a computer system and the operation of a computer?
3. What are the features of the various computer generations? Elaborate.
4. How the computers in the second-generation differed from the computers in the third generation?
5. Explain the different I/O devices used in a computer system? Why I/O devices are necessary for a computer system?
6. Why I/O devices are very slow as compared to the speed of primary storage and CPU?



Further Readings

Fundamental Computer Concepts, William S. Davis.

Fundamental Computer Skills, Feng-Qi Lai, David R. Hofmeister



Web Links

<http://books.google.co.in/bkshp?hl=en>